1. This week's Problem of the Week in Math is described as follows:

    There are thirty positive integers less than 100 that share a certain property. Your friend, Blake, wrote them down in the table to the left. But Blake made a mistake! One of the numbers listed is wrong and should be replaced with another. Which number is incorrect, what should it be replaced with, and why?

    The numbers are listed below.

6	10	14	15	21
22	26	33	34	35
38	39	46	51	55
57	58	62	65	69
75	77	82	85	86
87	91	93	94	95

    Use the fact that the "certain" property is that these numbers are all supposed to be the product of *unique* prime numbers to find and fix the mistake that Blake made.
    Reminder: Code your solution in an `R` script and copy it over to this `.Rnw` file.
    Hint: You may find the `%in%` operator and the `setdiff()` function to be helpful.

    Solution: I installed the Primes package (Keyes and Egeler, 2025) to use for creating a list of my prime numbers. From there I created a loop that multiplied each prime number together ensuring that i and j did not equal each other and that the product was less than 100. After storing those products in an empty vector I compared that to the original table that we were given using the `setdiff` function to compare the vectors and replace the wrong number which was 75 with 74. 75 is wrong because its prime factorization is 3*5*5 which includes a repeated prime number violating our unique prime product rule. 74, the replacement, is a unique prime number because its prime factors are 2*37 making it fit our given conidition.

```r
install.packages("primes", repos= "https://cran.rstudio.com/")

## package 'primes' successfully unpacked and MD5 sums checked
##
## The downloaded binary packages are in
##  C:\Users\schne\AppData\Local\Temp\RtmpOIZhQb\downloaded_packages

library(primes)

prime.numbers <- generate_primes(max = 100) #this creates our vector of prime number that we will be using



prime.products <- c() #this is our empty vector that stores our prime products created from the loop

for (i in 1:length(prime.numbers)) {
  for (j in 1:length(prime.numbers)) {
    if( i != j){
      product <- prime.numbers[i] * prime.numbers[j]

      if (!is.na(product) && product < 100) {
        prime.products <- c(prime.products, product)
      }
    }
  }
}
prime.products <- unique(sort(prime.products))
#used this to sort my products and make sure I didn't have an duplicates
#I also used the print function in my console with the unique prime products as a gut to make sure my loop worked

given.numbers <- c( 6, 10, 14, 15, 21,
                    22, 26, 33, 34, 35,
                    38, 39, 46, 51, 55,
                    57, 58, 62, 65, 69,
                    75, 77, 82, 85, 86,
                    87, 91, 93, 94, 95)
```

```r
incorrect_number <- setdiff(given.numbers, prime.products)

correct_number <- setdiff(prime.products, given.numbers)

cat("Incorrect Number in Table:", incorrect_number, "\n")

## Incorrect Number in Table: 75

cat("Correct Number to Replace It:", correct_number, "\n")

## Correct Number to Replace It: 74

#this cat function is used to print the incorrect and correct number
```

References

Keyes, O. and Egeler, P. (2025). *primes: Fast Functions for Prime Numbers*. R package version 1.6.1.